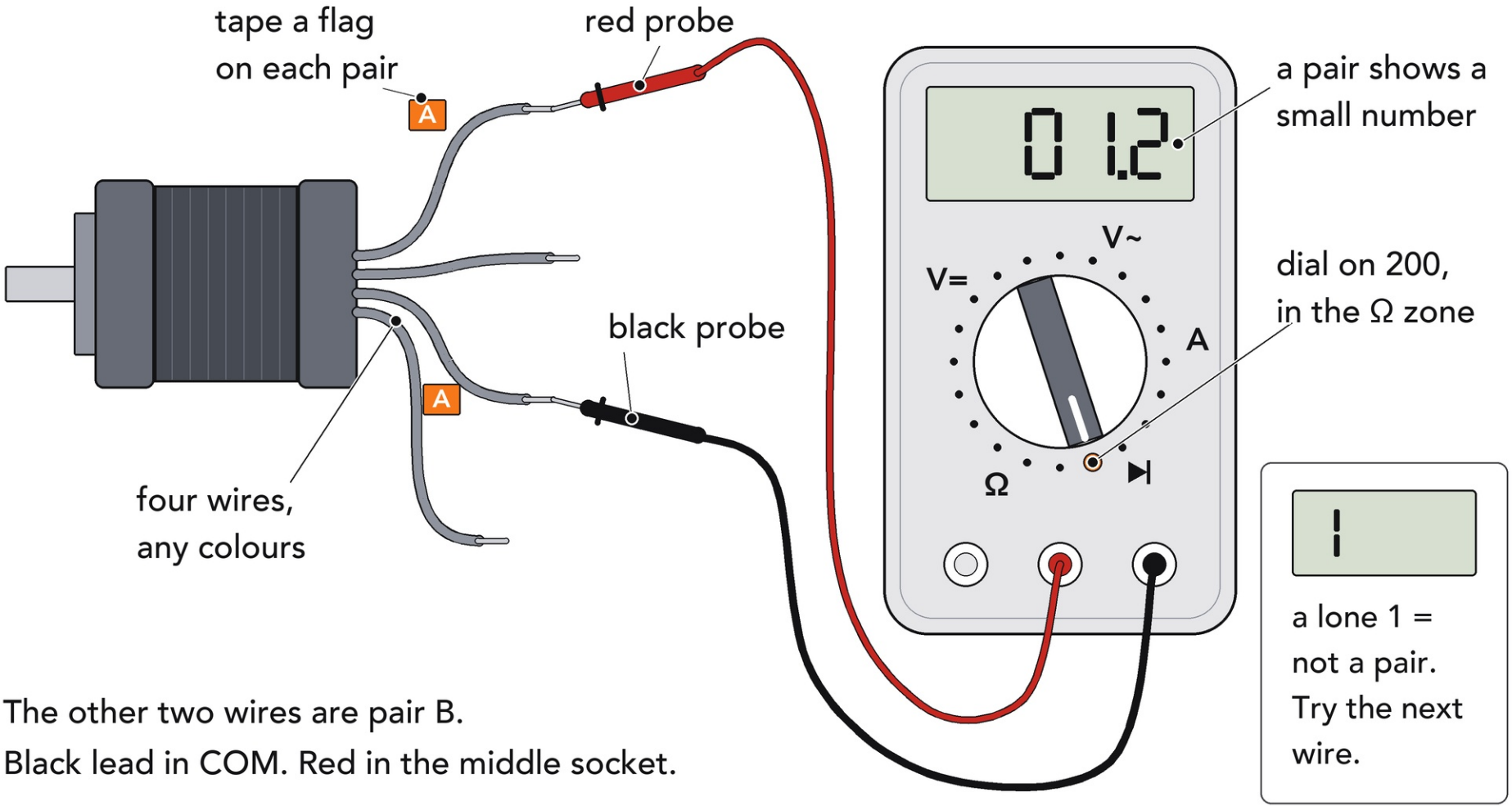
Step E1 Find the two coil pairs with the meter.

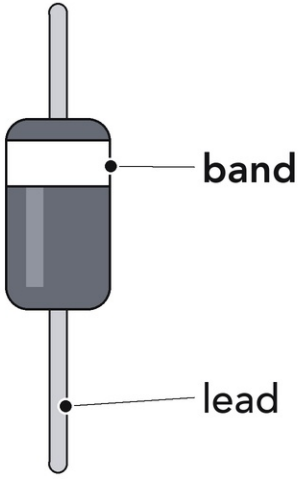


The other two wires are pair B.

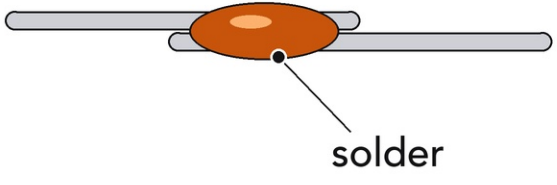
Black lead in COM. Red in the middle socket.

Step E2 Build bridge A. Four diodes, every band up.

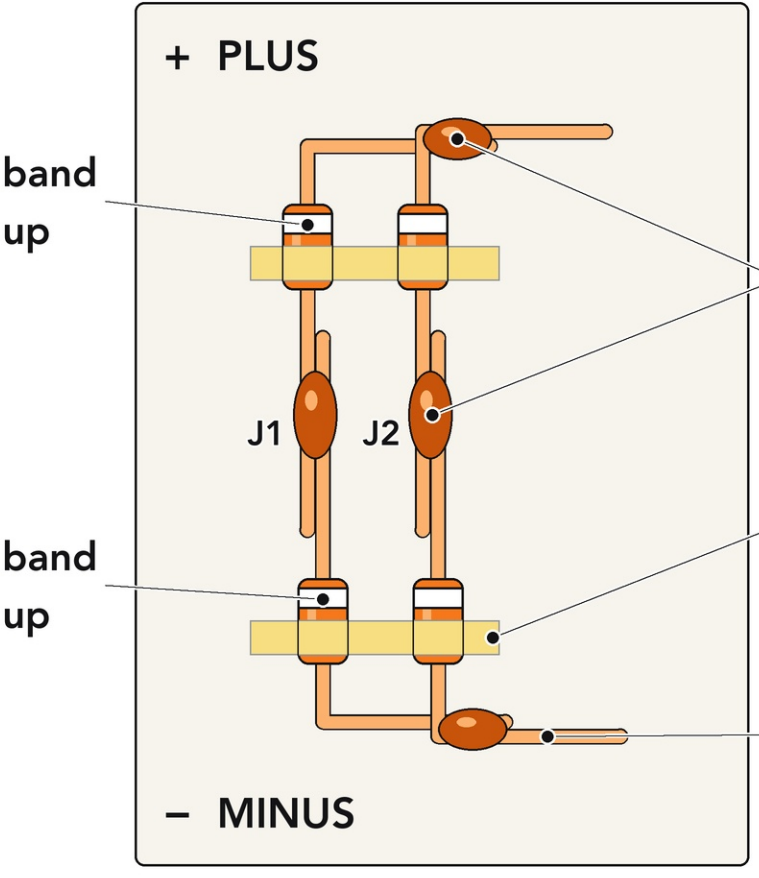
1. this ring is the band



2. two leads side by side,
2 cm. Solder the middle.



3. tape them on the card, bands toward PLUS

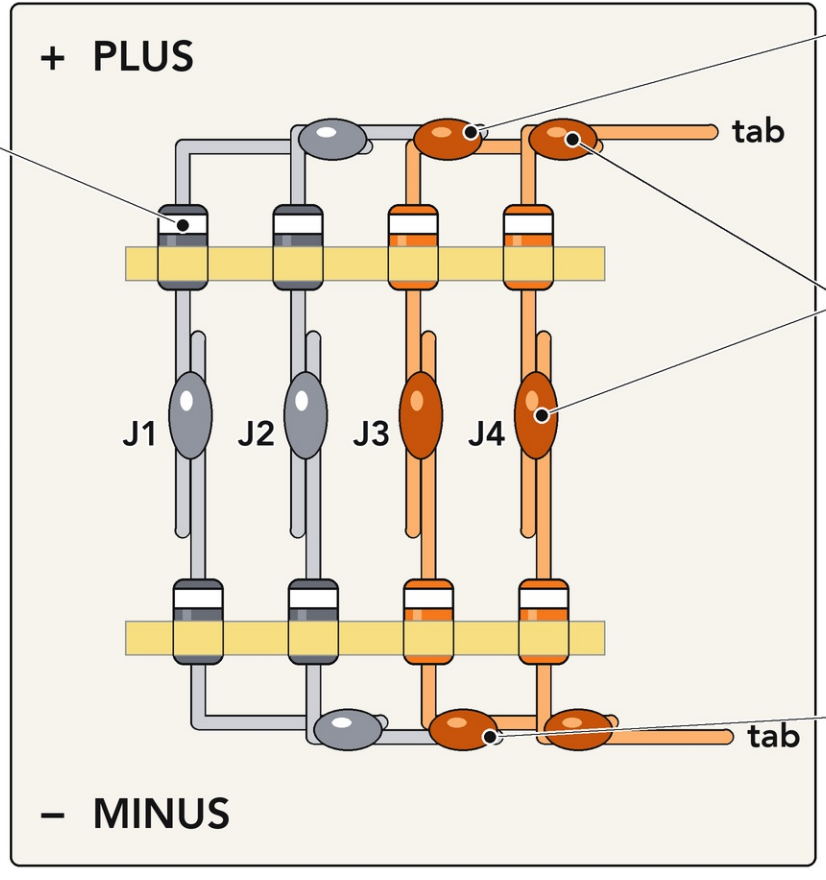


4. solder four joints

tape

bend each end lead over a screwdriver

Step E3 Add bridge B. Join plus to plus, minus to minus.



8 bands, all pointing at PLUS

the top row of bent leads is the PLUS rail

six new joints

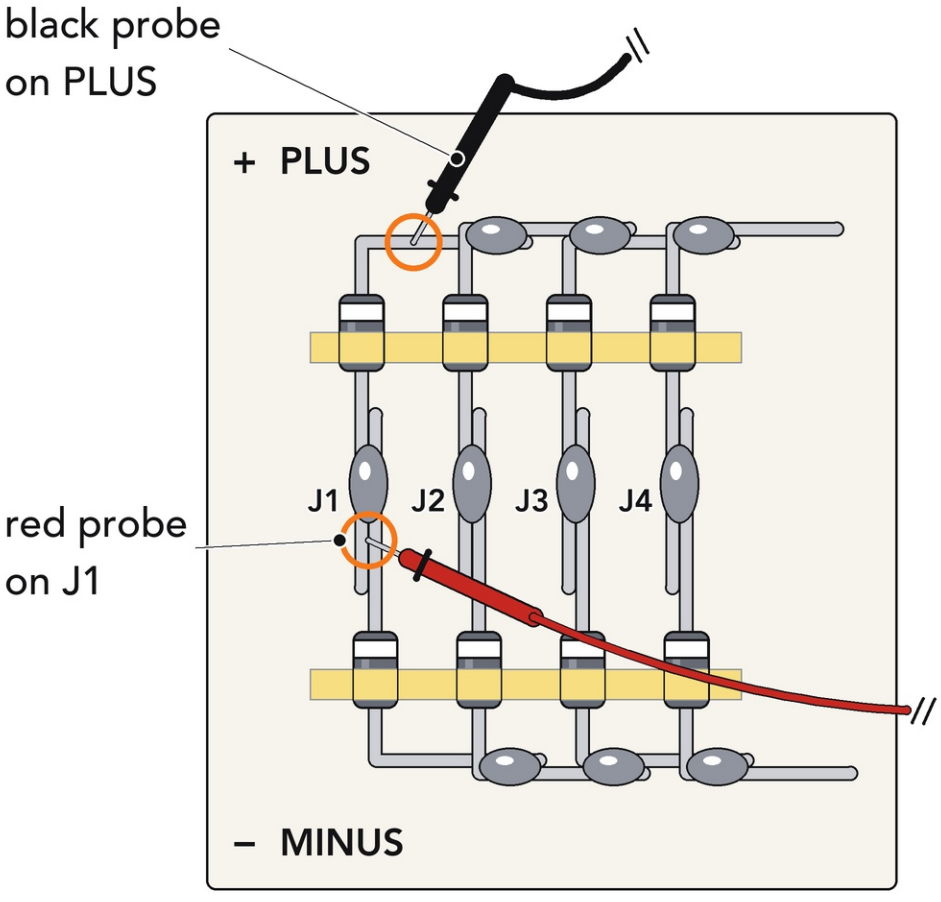
leave the two tabs bare

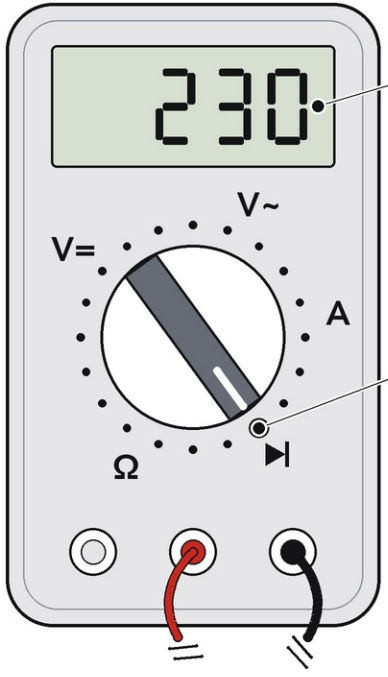
the bottom row is the MINUS rail



a band pointing down is wrong

Step E4 Meter check: red on J, black on plus.

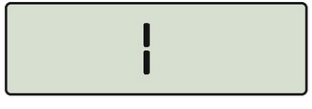




70 to 350
= good

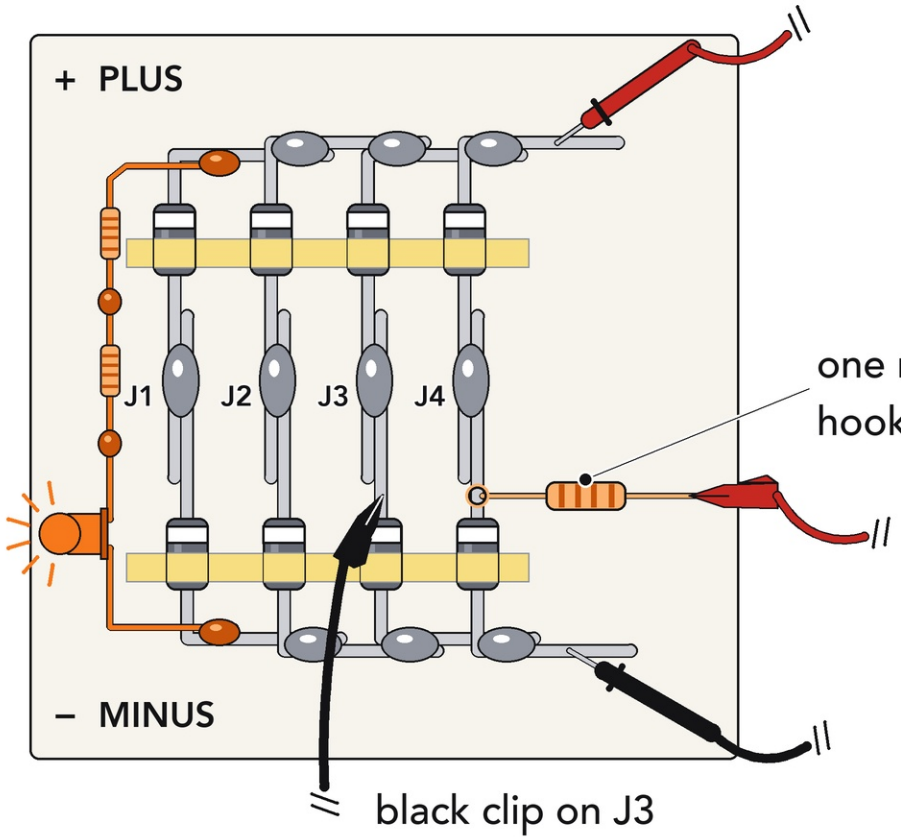
dial on the
diode mark

Then J2, J3 and J4 the same way.
Then black on each J, red on the MINUS rail.



a lone 1 here: that
diode is backwards

Step E5 Add the light. First power through one 1 k.



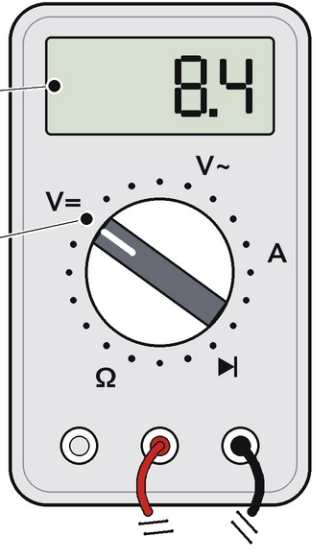
one more 1 k,
hooked on J4

black clip on J3

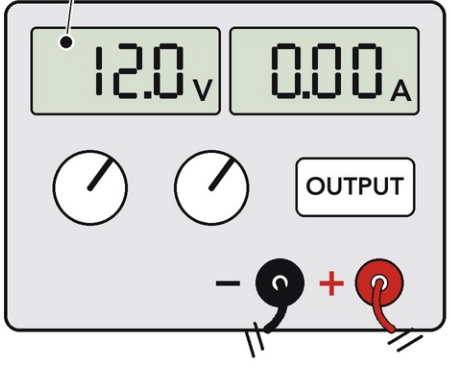
The light: two 1 k and the LED in one string.
Resistor end on PLUS. LED short leg on MINUS.

8 to 9 V = good

dial on 20,
in the V= zone



12.0 V

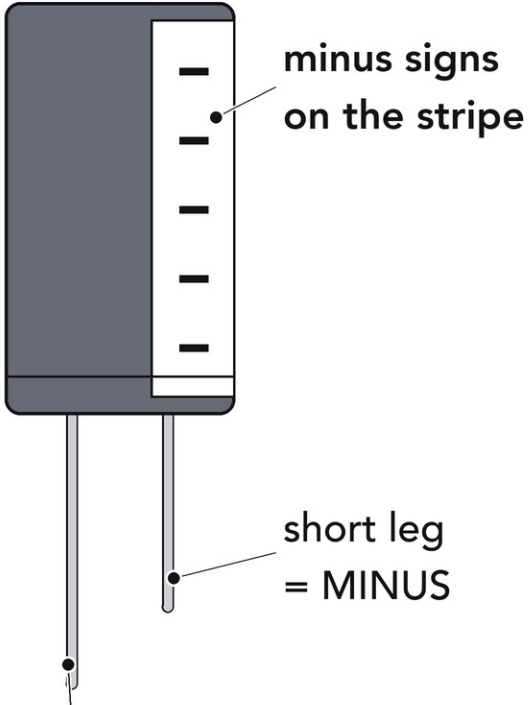


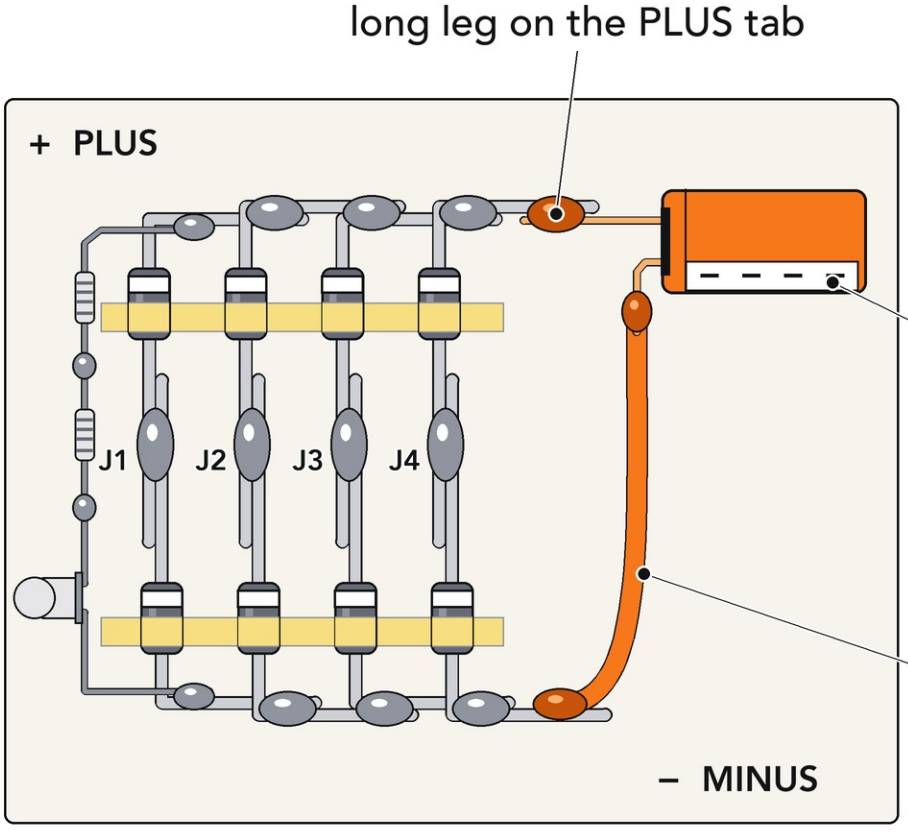
Under 3 V, or a minus sign: stop.

Then swap the two clips. Then do J1 and J2.

Step E6 Add the capacitor. Stripe side to minus.

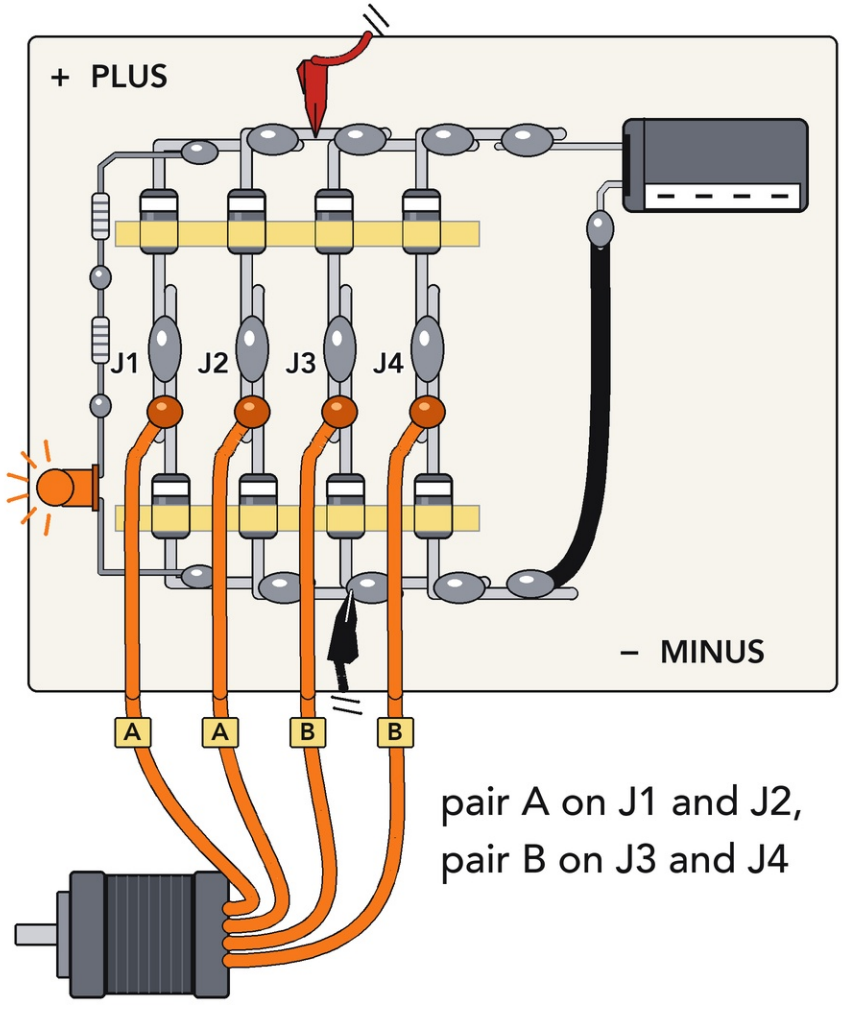
1000 uF 50 V





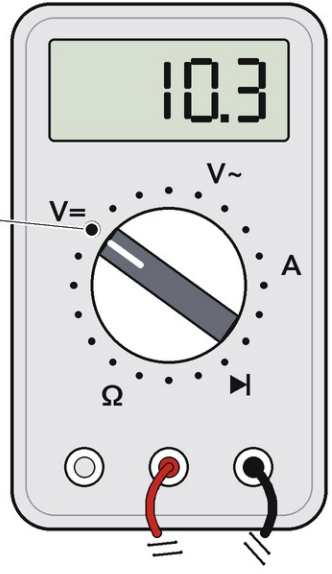
Then first power again through the 1 k (picture E5).
Climbs to 8 to 9 V in 3 seconds = good.
Stays under 3 V = capacitor backwards. Fit a new one.

Step E7 Solder the motor wires on. Crank with no load.



pair A on J1 and J2,
pair B on J3 and J4

dial on 20,
in the V= zone

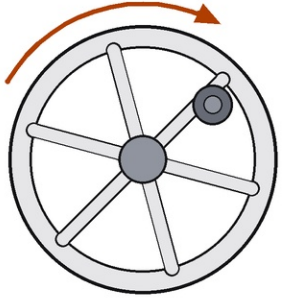


30 turns a minute:
4 to 6 V

60 turns a minute:
9 to 13 V

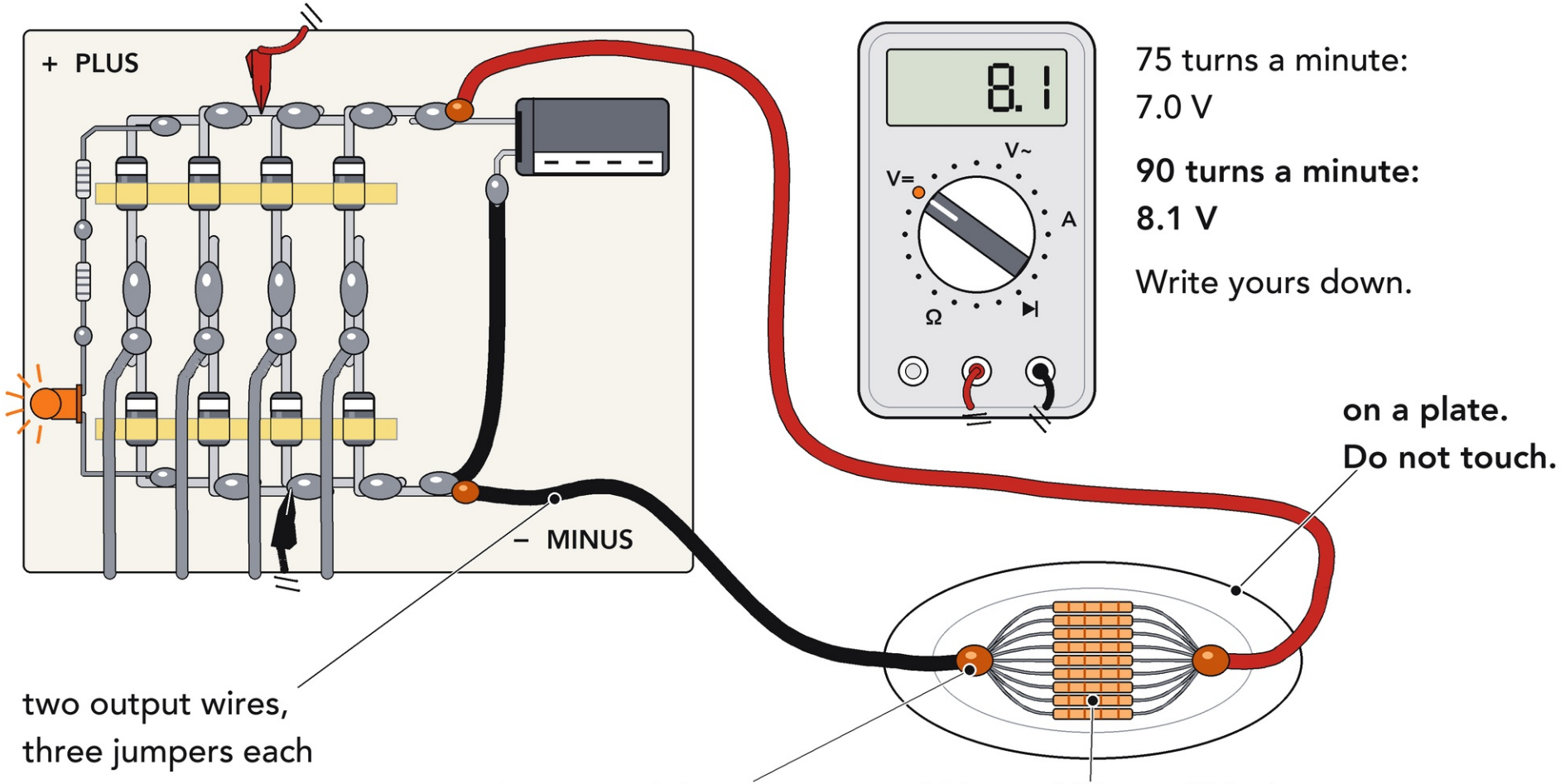
90 turns a minute:
13 to 20 V
(dial on 200)

Nothing plugged in? Never faster than 2 turns a second.



one turn a second
= 60 turns a minute

Step E8 Crank into thirty kit resistors. Ten seconds only.



75 turns a minute:
7.0 V

90 turns a minute:
8.1 V

Write yours down.

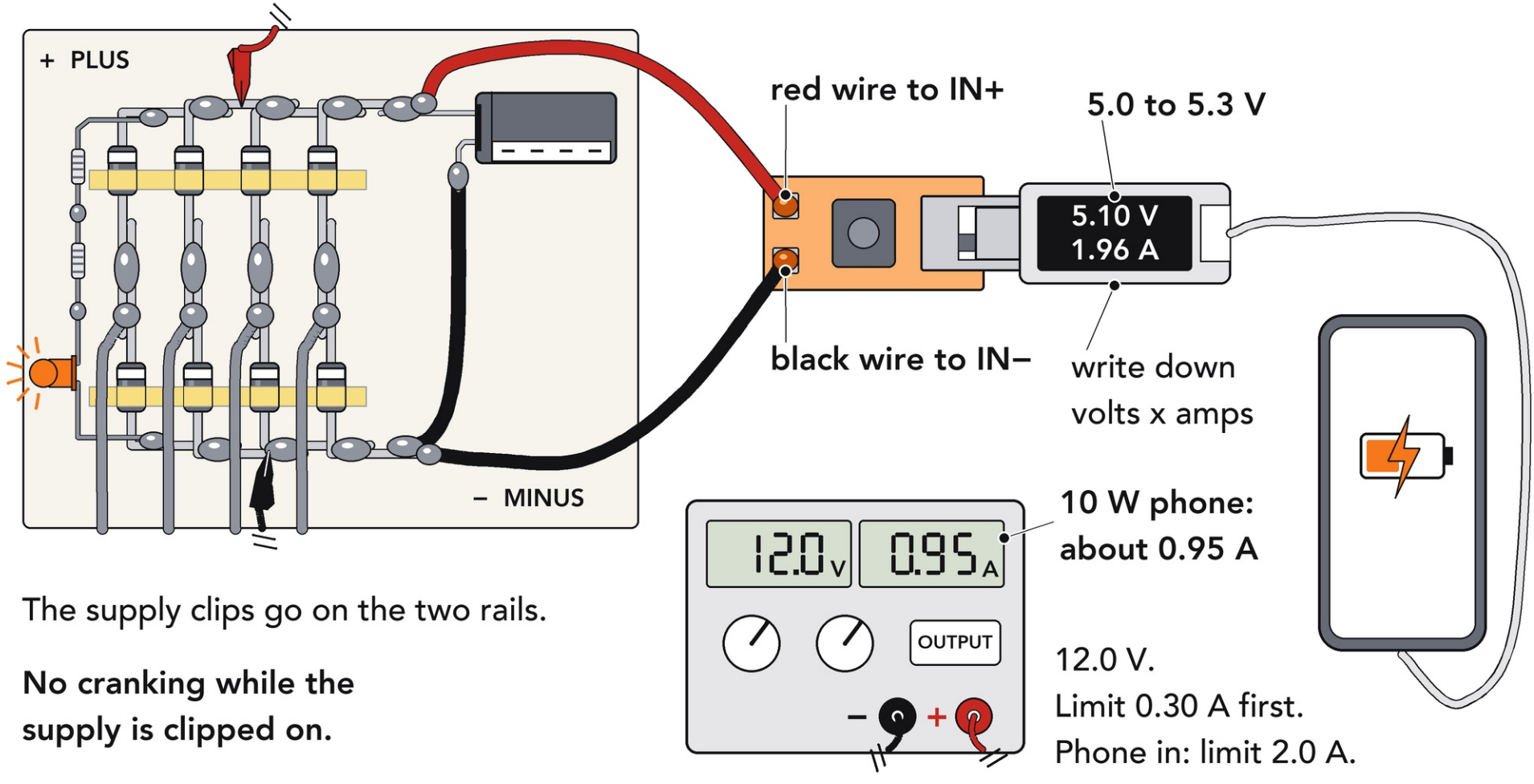
on a plate.
Do not touch.

two output wires,
three jumpers each

one wire on each lump,
either way round

ten 100, ten 220, ten 330 ohm,
twisted together at each end

Step E9 Wire the module in. Test it on the supply.



The supply clips go on the two rails.

No cranking while the supply is clipped on.

Step E10 The film: crank, tester in shot, sixty seconds.

